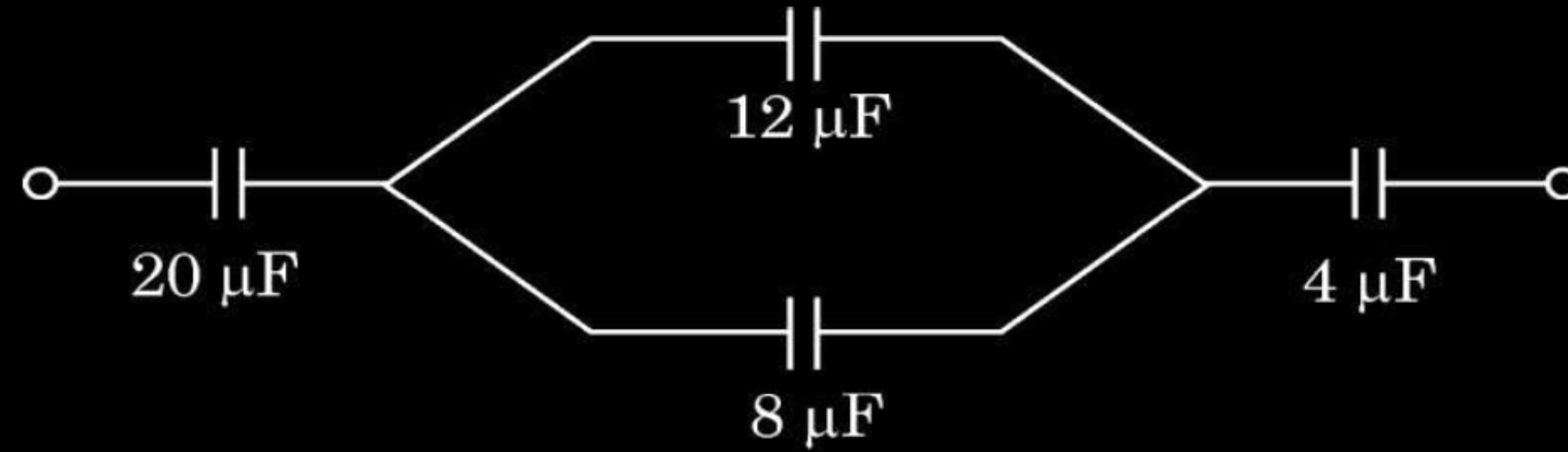


1. In the circuit shown, the charge on the left plate of $20\ \mu\text{F}$ capacitor is $-50\ \mu\text{C}$. The charge on the right plate of the $12\ \mu\text{F}$ capacitor is :



- | | |
|-----------------------|------------------------|
| (A) $20\ \mu\text{C}$ | (B) $-20\ \mu\text{C}$ |
| (C) $30\ \mu\text{C}$ | (D) $-30\ \mu\text{C}$ |

- 33.** (a) (i) A parallel plate capacitor with plate area A and plate separation d has a capacitance C_0 . A slab of dielectric constant K having area A and thickness $\left(\frac{d}{4}\right)$ is inserted in the capacitor, parallel to the plates. Find the new value of its capacitance.
- (ii) You are provided with a large number of $1\ \mu\text{F}$ identical capacitors and a power supply of $1200\ \text{V}$. The dielectric medium used in each capacitor can withstand up to $200\ \text{V}$ only. Find the minimum number of capacitors and their arrangement, required to build a capacitor system of equivalent capacitance of $2\ \mu\text{F}$ for use with this supply.

- (b) (i) An electric dipole of dipole moment p consists of point charges q and $-q$, separated by $2a$. Derive an expression for electric potential in terms of its dipole moment at a point at a distance x ($\gg a$) from its centre and lying (I) along its axis, and (II) along its bisector line.
- (ii) An electric dipole of dipole moment $\vec{p} = (0.8\hat{i} + 0.6\hat{j}) 10^{-29} \text{ Cm}$ is placed in an electric field $\vec{E} = 1.0 \times 10^7 \hat{k} \frac{\text{V}}{\text{m}}$. Calculate the magnitude of the torque acting on it and the angle it makes with the x-axis, at this instant.

1. The electric field (E) and electric potential (V) at a point inside a charged hollow metallic sphere are respectively :

(A) $E = 0, V = 0$

(B) $E = 0, V = V_0$ (a constant)

(C) $E \neq 0, V \neq 0$

(D) $E = E_0$ (a constant), $V = 0$

28. Two point charges of $-5 \mu\text{C}$ and $2 \mu\text{C}$ are located in free space at $(-4 \text{ cm}, 0)$ and $(6 \text{ cm}, 0)$ respectively.

(a) Calculate the amount of work done to separate the two charges at infinite distance.

(b) If this system of charges was initially kept in an electric field $E = \frac{A}{r^2}$, where $A = 8 \times 10^4 \text{ N C}^{-1} \text{ m}^2$, calculate the electrostatic potential energy of the system.

1. Two horizontal plates, separated by 1 cm, are arranged one above the other. A particle of mass 5 mg and charge 2 nC is released in air between the plates. The potential difference that should be applied to the plates so that the particle remains suspended between them, is :

(A) 250 V

(B) 200 V

(C) 100 V

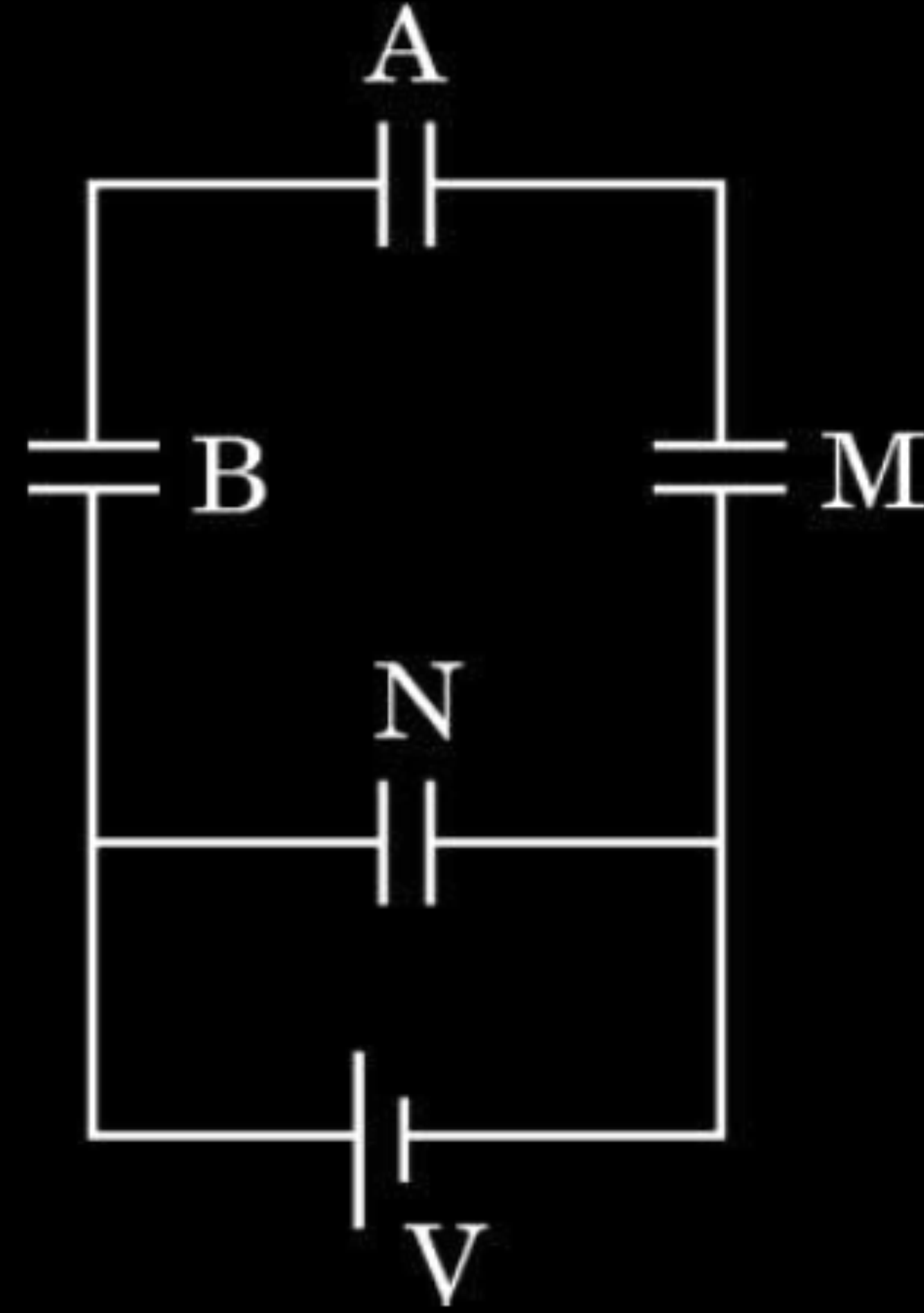
(D) 50 V

29. A capacitor is a system of two conductors separated by an insulator. In practice, the two conductors have charges Q and $-Q$ with potential difference $V = V_1 - V_2$ between them. The ratio $\frac{Q}{V}$ is a constant, denoted by C and is called the capacitance of the capacitor. It is independent of Q or V . It depends only on the geometrical configuration (shape, size, separation) of the two conductors and the medium separating the conductors. When a parallel plate capacitor is charged, the electric field E_0 is localised between the plates and is uniform throughout. When a slab of a dielectric is inserted between the charged plates (charge density σ), the dielectric is polarised by the field. Consequently opposite charges appear on the faces of the slab, near the plates, with surface charge density of magnitude σ_p . For a linear dielectric σ_p is proportional to E_0 . Introduction of a dielectric changes the electric field, and hence, the capacitance of a capacitor, and hence, the energy stored in the capacitor.

Like resistors, capacitors can also be arranged in series or in parallel or in a combination of series and parallel.

- (i) (a) Three capacitors A, B and M, each of capacitance C are connected to a capacitor N of capacitance $2C$ and a battery as shown in the figure. If the charges on A and N are Q and Q' respectively, then $\frac{Q'}{Q}$ is :

1



(A) $\frac{1}{6}$

(B) $\frac{1}{3}$

(C) 3

(D) 6

(b) A slab (area A and thickness $\frac{d}{2}$) of dielectric constant K is inserted in a parallel plate capacitor of plate area A and plate separation d . If C and C_0 are the capacitances of the capacitors with and without the dielectric, then $\frac{C}{C_0}$ is : *1*

(A) $\frac{K+1}{2K}$

(B) $\frac{2K}{K+1}$

(C) $\frac{K}{K-1}$

(D) $\frac{K-1}{K}$

- (ii) An electric field E is established between the plates of an air filled parallel plate capacitor, with charges Q and $-Q$. V is the volume of the space enclosed between the plates. The energy stored in the capacitor is :

(A) $\frac{1}{2} \epsilon_0 E^2$

(B) $\epsilon_0 Q^2 E$

(C) $\frac{1}{2} \epsilon_0 E^2 V$

(D) $\epsilon_0 E Q V$

- (iii) A slab (area A and thickness d_1) of a linear dielectric of dielectric constant K is inserted between charged plates (charge density σ) of a parallel plate capacitor [plate area A and plate separation d ($> d_1$)] and opposite charges with charge density of magnitude σ_p appear on the faces of the slab. The dielectric constant K is given by : 1

(A) $\frac{\sigma + \sigma_p}{\sigma}$

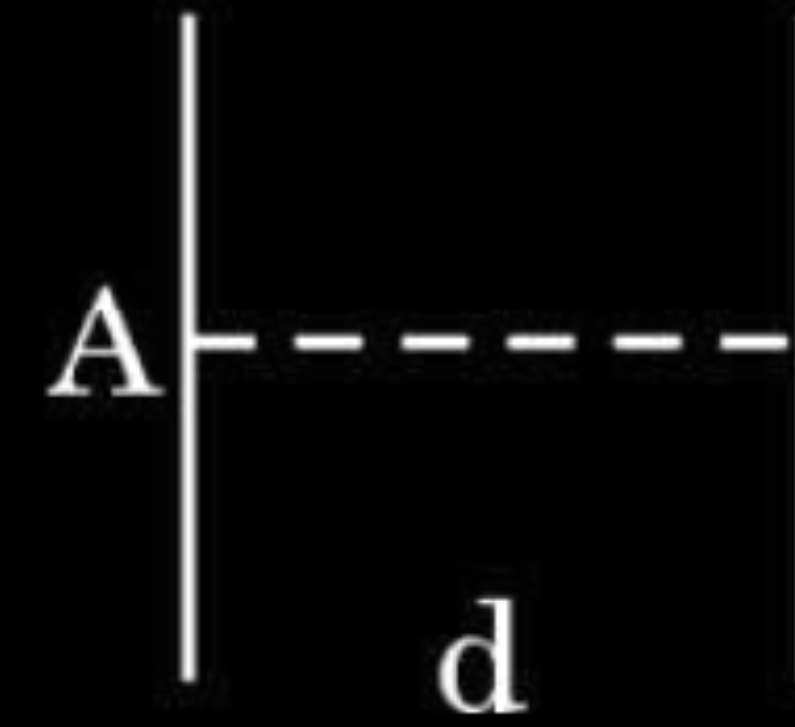
(B) $\frac{\sigma}{\sigma - \sigma_p}$

(C) $\frac{\sigma + \sigma_p}{\sigma_p}$

(D) $\frac{\sigma}{\sigma_p}$

- (iv) Consider a capacitor of capacitance C , with plate area A and plate separation d , filled with air [Fig. (a)]. The distance between the plates is increased to $2d$ and one of the plates is shifted as shown in Fig. (b). The capacitance of the new system now is :

1



(a)



(b)

(A) $\frac{C}{4}$

(B) $\frac{C}{2}$

(C) $2C$

(D) $4C$

- 32.** (a) (i) The electric field in a region is given by $\vec{E} = 40x \hat{i}$ N/C. Find the amount of work done in taking a unit positive charge from a point (0, 3m) to the point (5m, 0).
- (ii) A charge Q is distributed over two concentric hollow spheres of radii r and R ($> r$) such that their surface charge densities are equal. Find :
- (I) the electric field, and
- (II) the potential at their common centre.

29. A power supply is connected across the two plates of a capacitor. Consequently, one plate becomes positively charged and the other plate becomes negatively charged. Thus a uniform electric field is established between the two plates. If a charged particle is released between these two plates, it experiences a force. The potential difference between the plates affects its charge and the energy stored in the capacitor. $4 \times 1 = 4$

(i) A proton and an electron are released from rest in the space between the two plates of a charged capacitor. Assume that the proton and the electron do not interact with each other. The acceleration of the proton compared with that of the electron is

1

- (A) same in both magnitude and direction.
- (B) same in magnitude but in opposite direction.
- (C) greater in magnitude and opposite in direction.
- (D) lesser in magnitude but in opposite direction.

(ii) An electron is released between the two parallel plates of a charged capacitor. Which of the following statements is incorrect ?

1

- (A) The electric field between the plates is directed from the positively charged plate to the negatively charged plate.
- (B) The magnitude of electric field is the same at all points between the plates.
- (C) The acceleration of the electrons will be constant between the plates.
- (D) The potential increases in the direction of the electric field between the plates.

- (iii) The potential difference between the two plates of a capacitor is V . An electron of mass m and charge $-e$ is released from rest near the negative plate. The maximum speed gained by the electron is **1**

(A) $2eVm$ (B) $\sqrt{\frac{em}{2V}}$ (C) $\sqrt{\frac{2eV}{m}}$ (D) $\sqrt{\frac{2me}{V}}$

- (iv) (a) The electric field between two parallel plates of a charged capacitor is 200 V/m . The plates are 5 mm apart. The potential difference between the two plates is **1**

(A) 20 V (B) 10 V
(C) 5.0 V (D) 1.0 V

OR

- (b) A proton is located at a point between the above plates. The force acting on the proton is **1**
- (A) $1.6 \times 10^{-19} \text{ N}$ (B) $2 \times 10^9 \text{ N}$
(C) $3.2 \times 10^{-17} \text{ N}$ (D) $4 \times 10^9 \text{ N}$

31. (a) (A) Two point charges q_1 and q_2 in air are located at $\vec{r}_1$ and $\vec{r}_2$ respectively in a region of uniform external field $\vec{E}$. Obtain an expression for the electrostatic potential energy of the system.
- (B) The coulomb force between two point charges is 1.5×10^{-4} N. How will this force be affected in the following situations :
- (i) the distance between the charges is doubled.
 - (ii) the magnitude of each charge is doubled but the distance between the charges remains the same.

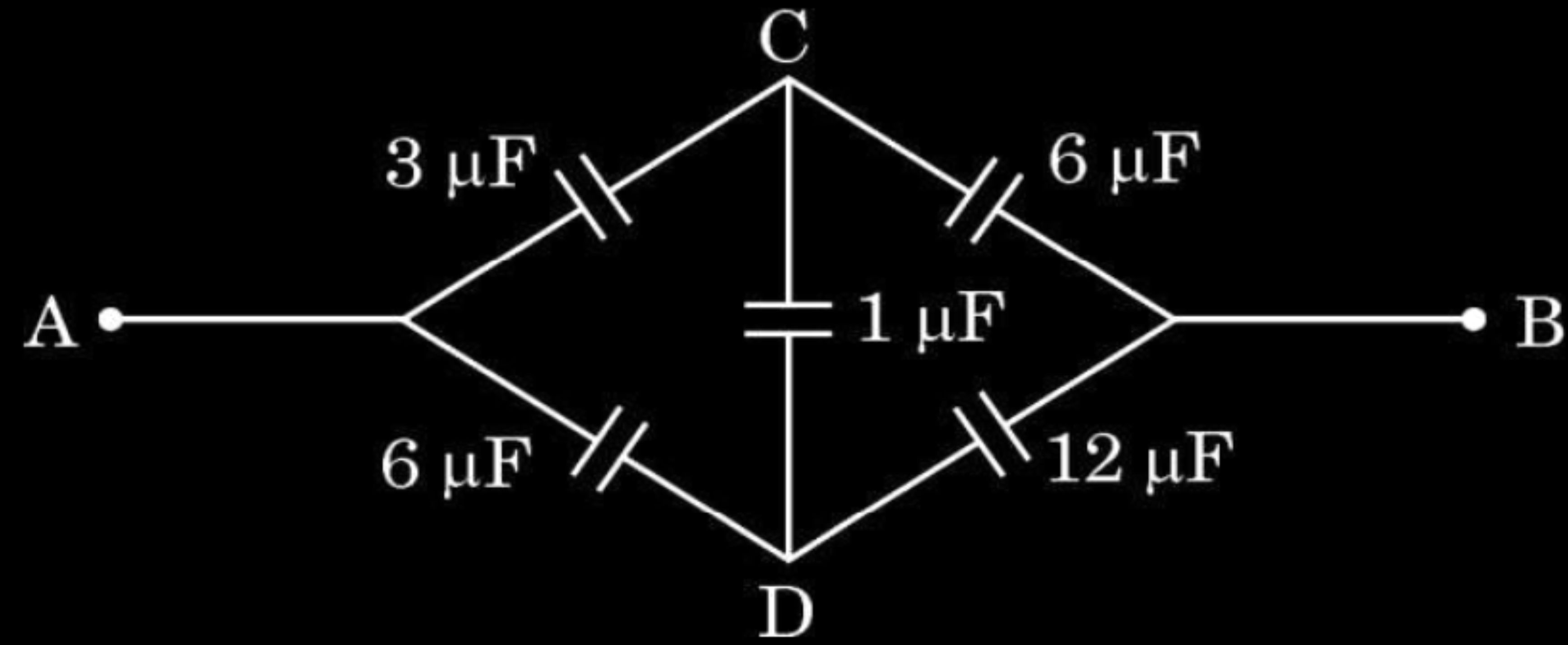
- (b) (A) Differentiate between the electrostatic force and the gravitational force between two charged particles.
- (B) A potential difference of 480 V is applied between two large horizontal plates, kept one above the other, 3.0 cm apart. A charged particle of mass 4×10^{-11} kg, when released at rest between the plates, remains stationary in the region. If the upper plate is at positive potential, find
- (i) nature of the charge on the particle
 - (ii) the magnitude of the charge on the particle.

- 22.** (a) “There is a limit to the amount of charge that can be stored on a given capacitor.” Explain.
- (b) A capacitor is charged by a battery to a potential difference V . It is disconnected from the battery and connected across another identical uncharged capacitor. Calculate the ratio of total energy stored in the combination to the initial energy stored in the capacitor.

4. A metal sheet is inserted between the plates of a parallel plate capacitor of capacitance C . If the sheet partly occupies the space between the plates, the capacitance :
- | | |
|---------------------------|------------------------------|
| (A) remains C | (B) becomes greater than C |
| (C) becomes less than C | (D) becomes zero |

- 26.** (a) Two small solid metal balls A and B of radii R and $2R$ having charge densities 2σ and 3σ respectively are kept far apart. Find the charge densities on A and B after they are connected by a conducting wire.

2. The effective capacitance of the network between points A and B shown in the figure is :



(A) $\frac{4}{3}\ \mu\text{F}$

(B) $2\ \mu\text{F}$

(C) $\frac{3}{2}\ \mu\text{F}$

(D) $6\ \mu\text{F}$

3. The electric field at a point in a region is given by $\vec{E} = \alpha \frac{\vec{r}}{|\vec{r}|^3}$, where α is a constant and r is the distance of the point from the origin. The magnitude of potential of the point is :

(A) $\frac{\alpha}{r}$	(B) $\frac{\alpha r^2}{2}$
(C) $\frac{\alpha}{2r^2}$	(D) $-\frac{\alpha}{r}$

3. Four point charges Q each, are held at the four corners of a square of side l . The amount of work done in bringing a charge Q from infinity to the centre of the square will be :

(A) $\frac{Q^2}{\pi \epsilon_0 l}$

(B) $\frac{\sqrt{2} Q^2}{\pi \epsilon_0 l}$

(C) $\frac{Q^2}{2\pi \epsilon_0 l}$

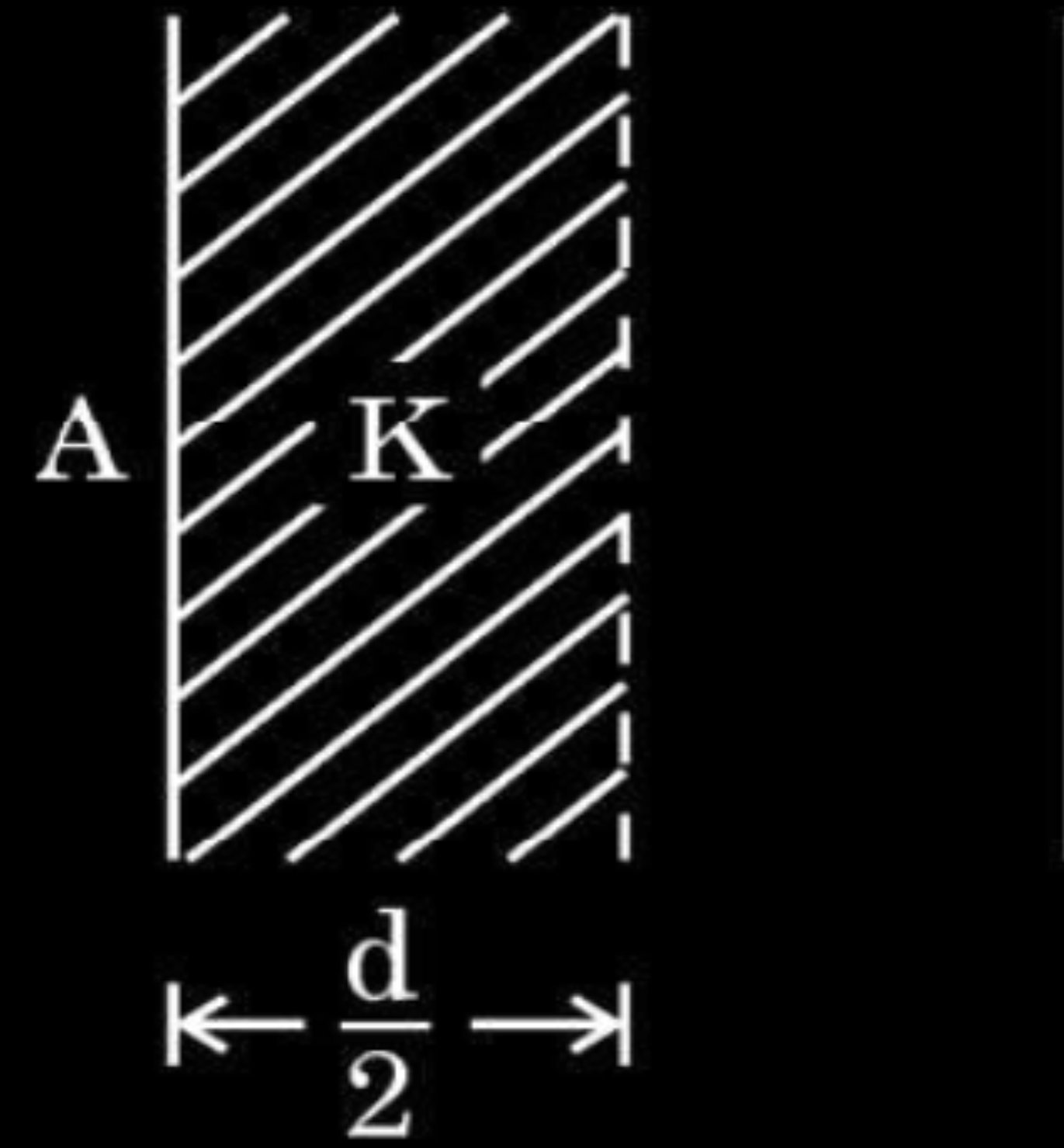
(D) Zero

24. A capacitor of plate area A and plate separation d is charged by a battery to voltage V . The battery is disconnected and plates are slowly pulled apart till the separation becomes $2d$. Find the value of :

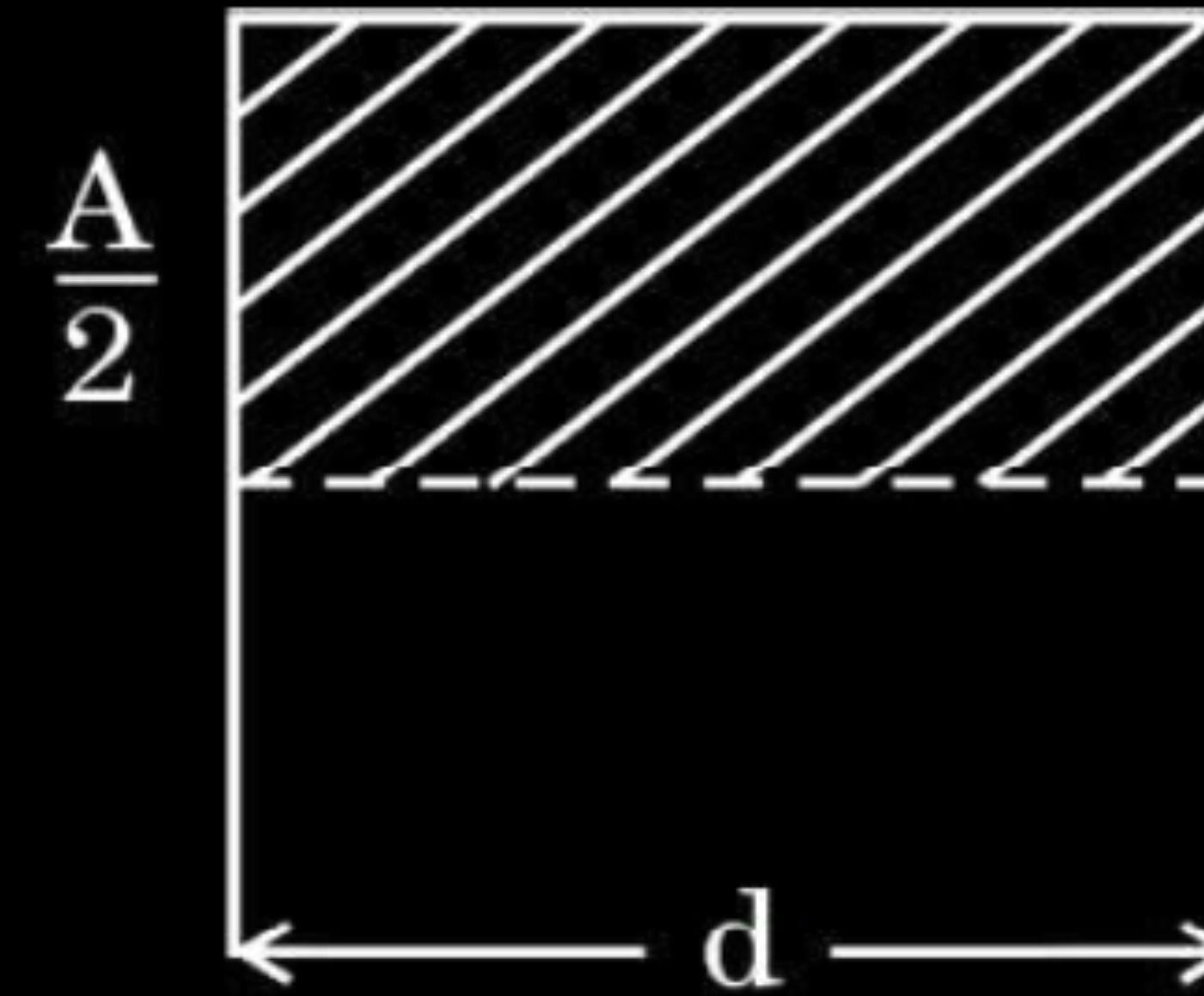
3

- (a) potential difference between the plates,
- (b) electric field between the plates,
- (c) work done in pulling the plates apart.

- 23.** A parallel plate capacitor has plate area A and plate separation d . Half of the space between the plates is filled with a material of dielectric constant K in two ways as shown in the figure.



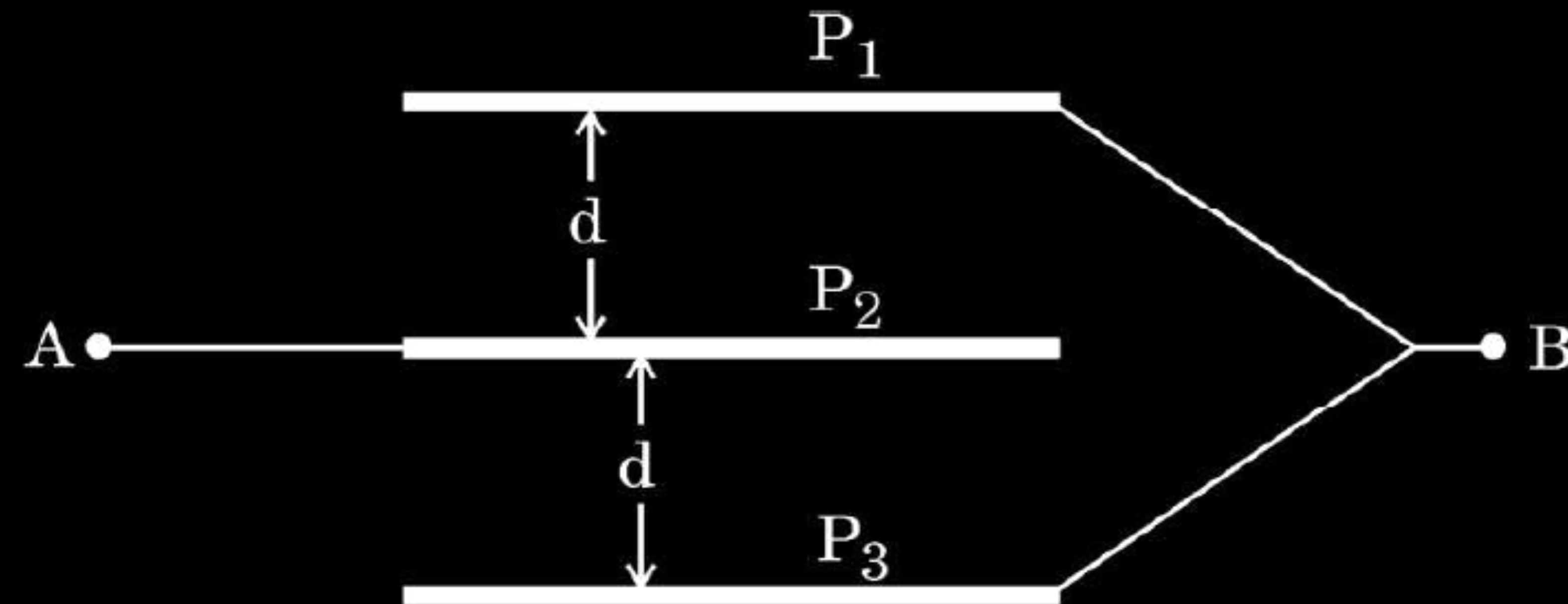
(a)



(b)

Find the values of the capacitance of the capacitors in the two cases.

30. A parallel plate capacitor consists of two conducting plates kept generally parallel to each other at a distance. When the capacitor is charged, the charge resides on the inner surfaces of the plates and an electric field is set up between them. Thus, electrostatic energy is stored in the capacitor. The figure shows three large square metallic plates, each of side 'L' held parallel and equidistant from each other. The space between P_1 and P_2 and P_2 and P_3 is completely filled with mica sheets of dielectric constant 'K'. The plate P_2 is connected to point A and other plates P_1 and P_3 are connected to point B. Point A is maintained at a positive potential with respect to point B and the potential difference between A and B is V.



- (i) The capacitance of the system between A and B will be :

(A) $\frac{\epsilon_0 K L^2}{d}$

(B) $\frac{\epsilon_0 K L^2}{2d}$

(C) $\frac{2\epsilon_0 K L^2}{d}$

(D) $\frac{2\epsilon_0 K d}{L^2}$

(ii) The charge on plate P_1 is : 1

(A) $\frac{\epsilon_0 V K L^2}{2d}$ (B) $\frac{\epsilon_0 V K L^2}{d}$

(C) $\frac{2\epsilon_0 V K L^2}{d}$ (D) $\frac{\epsilon_0 V K L^2}{4d}$

(iii) The electric field in the region between P_1 and P_2 is : 1

(A) $\frac{V}{d}$ (B) $\frac{2V}{d}$

(C) $\frac{V}{2d}$ (D) $\frac{d}{V}$

- (iv) (a) The separation between the plates of same area (L^2) of a parallel plate air capacitor having capacitance equal to that of this system, will be :

1

(A) $\frac{d}{K}$

(B) $\frac{2d}{K}$

(C) $\frac{d}{2K}$

(D) $\frac{d}{4K}$

OR

- (b) If the source of potential difference applied between A and B is removed, and then A and B are connected by a conducting wire, the net charge on the system will be :

1

(A) $\frac{\epsilon_0 V K L^2}{4d}$

(B) $\frac{\epsilon_0 V K L^2}{2d}$

(C) $\frac{\epsilon_0 V K L^2}{d}$

(D) Zero

- (ii) Three point charges of -2 nC , -1 nC , and $+5 \text{ nC}$ are kept at the vertices A, B and C of an equilateral triangle of side 0.2 m . Find the total amount of work done in shifting the charges from A to A_1 , B to B_1 and C to C_1 . Here A_1 , B_1 and C_1 are the midpoints of sides AB, BC and CA, respectively.

- (ii) A point charge of $+2\text{ nC}$ is kept at the origin of a three-dimensional coordinate system. Find the type and magnitude of the charge which should be kept at $(0, 0, -6\text{m})$ so that the potential due to the system becomes zero at $(0, 0, 2\text{m})$.

30. A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates. $4 \times 1 = 4$

(i) The electric field between the plates of a parallel plate capacitor is E . Now the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value. The new value of the electric field between the plates will be :

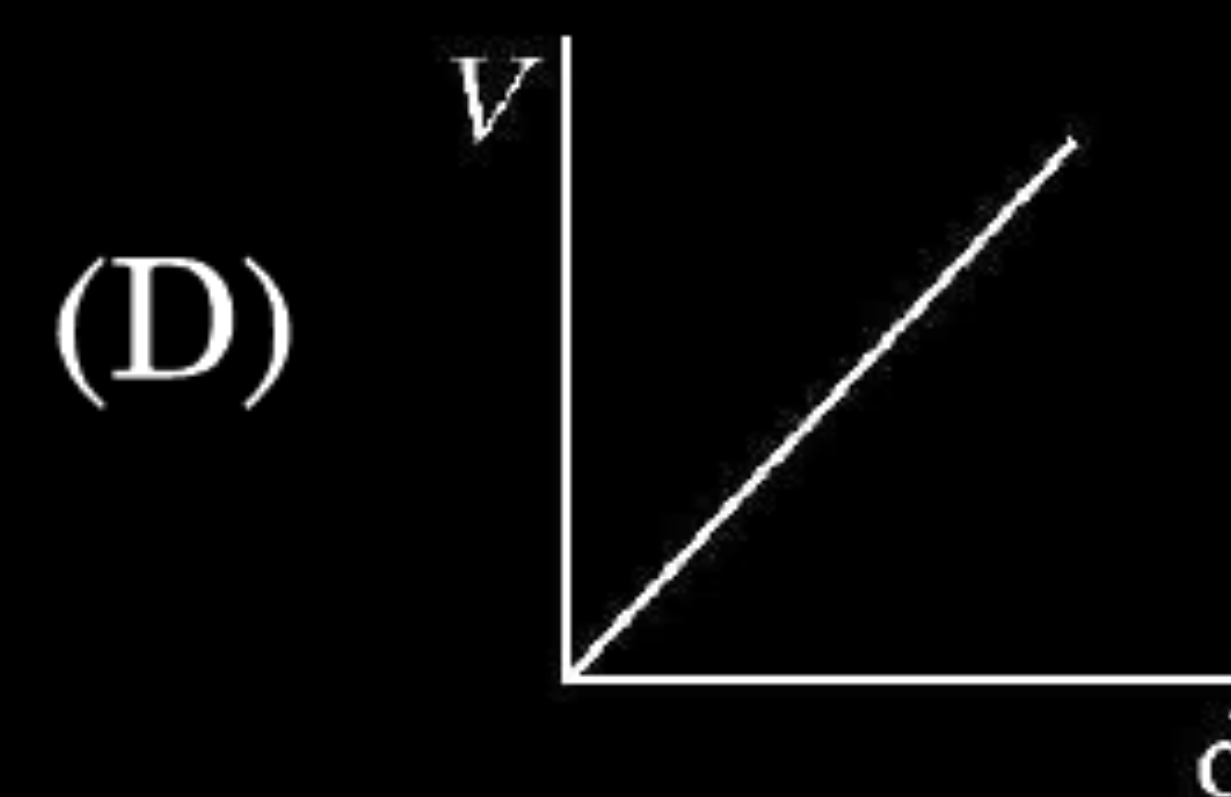
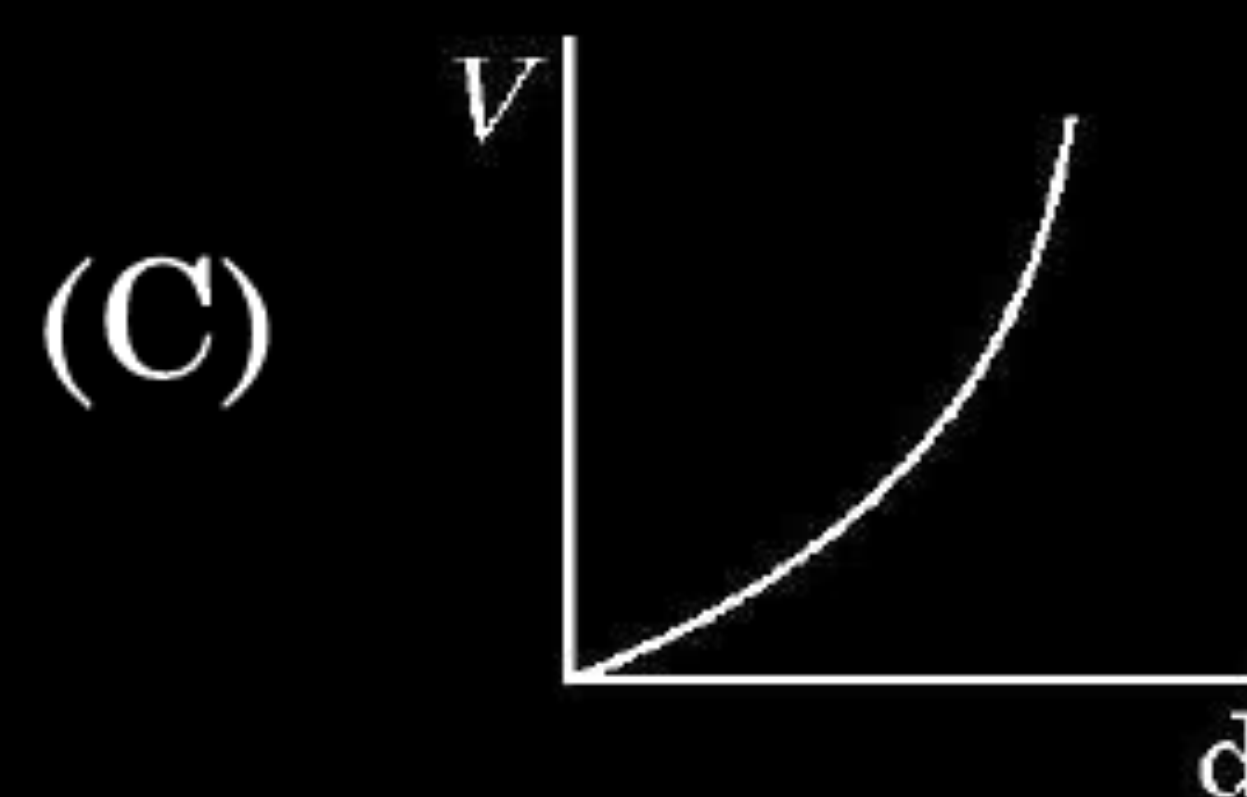
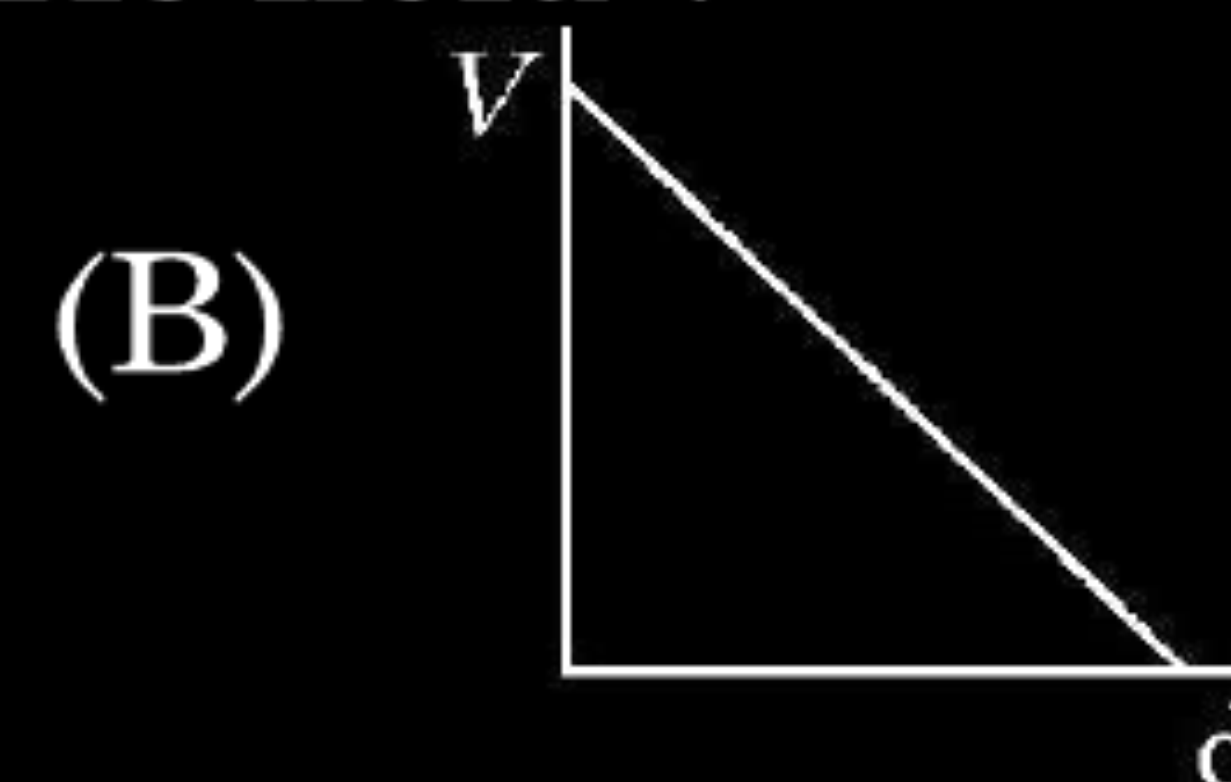
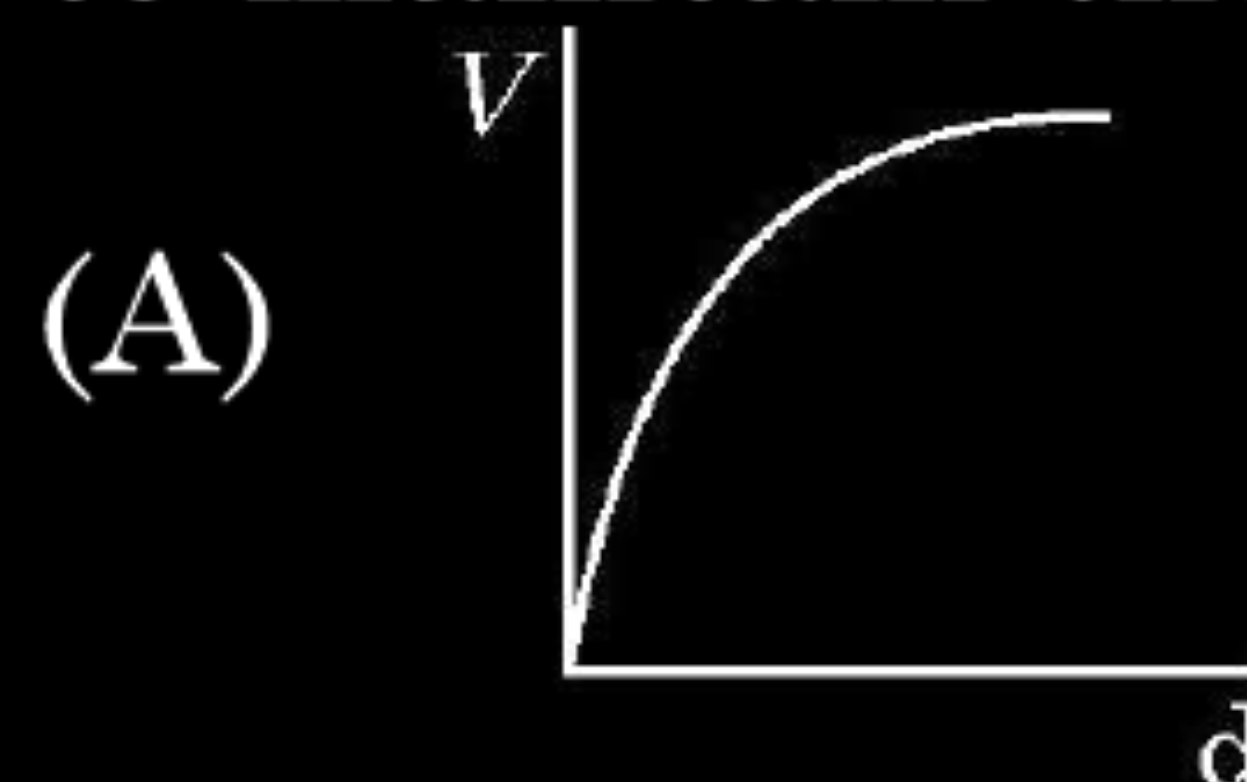
(A) E

(B) $2E$

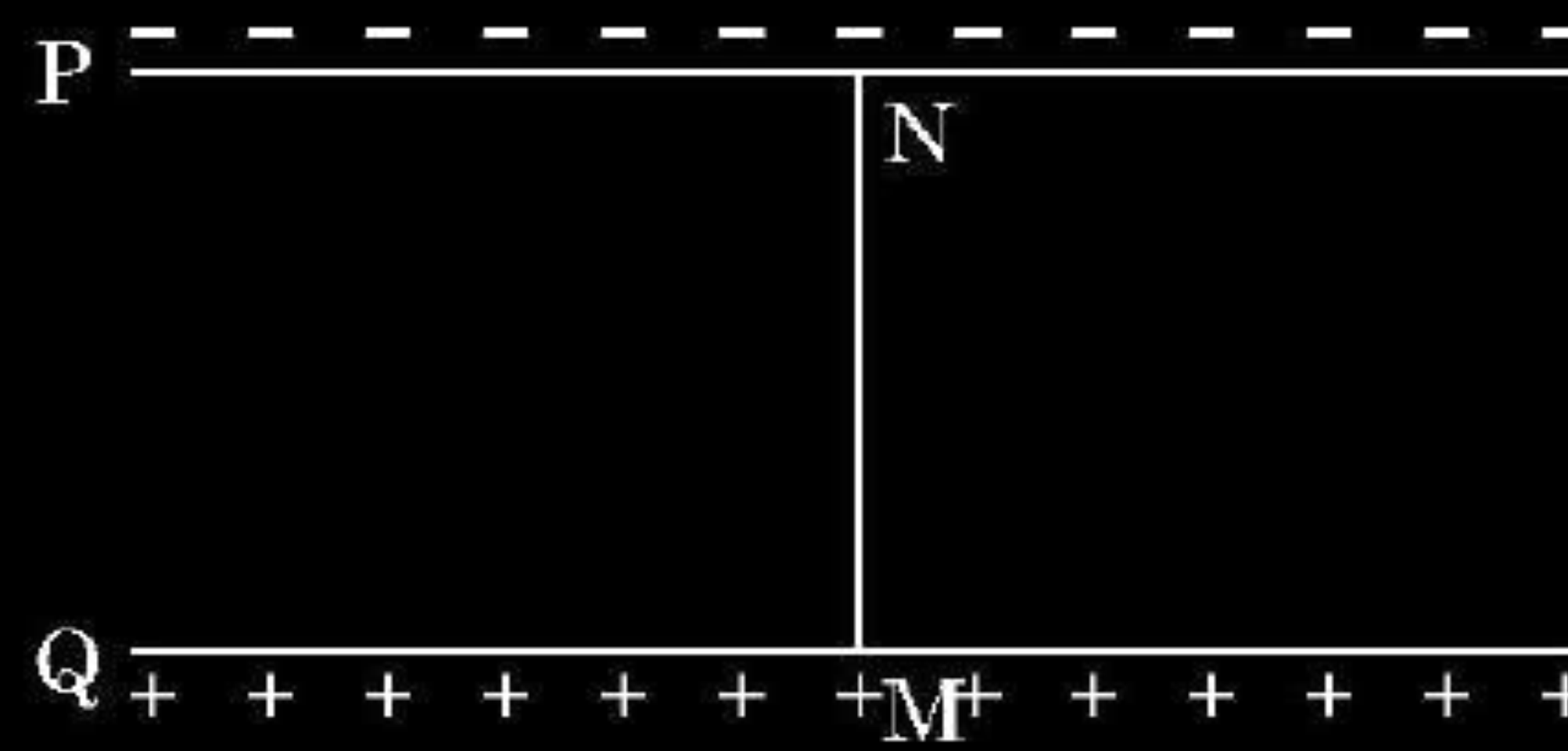
(C) $\frac{E}{4}$

(D) $\frac{E}{2}$

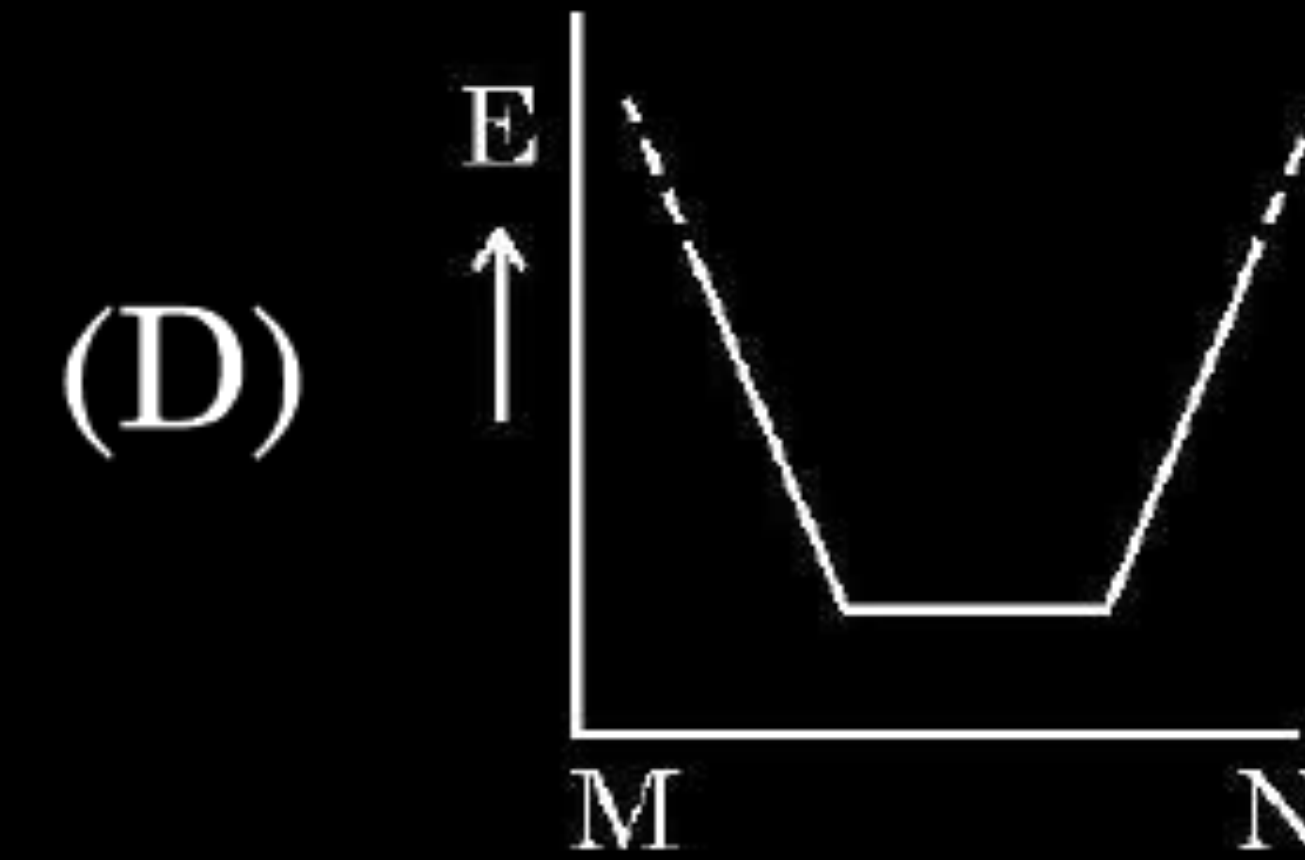
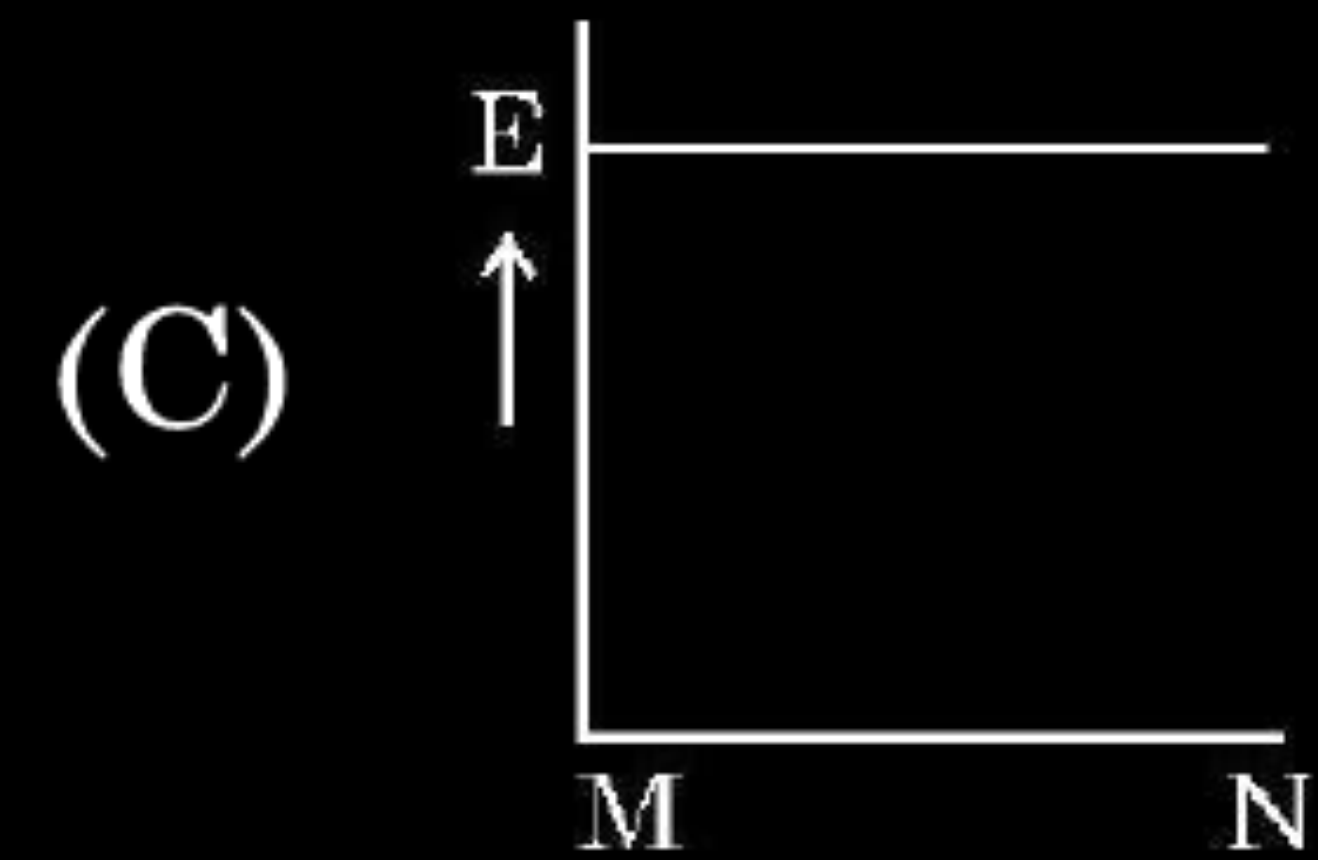
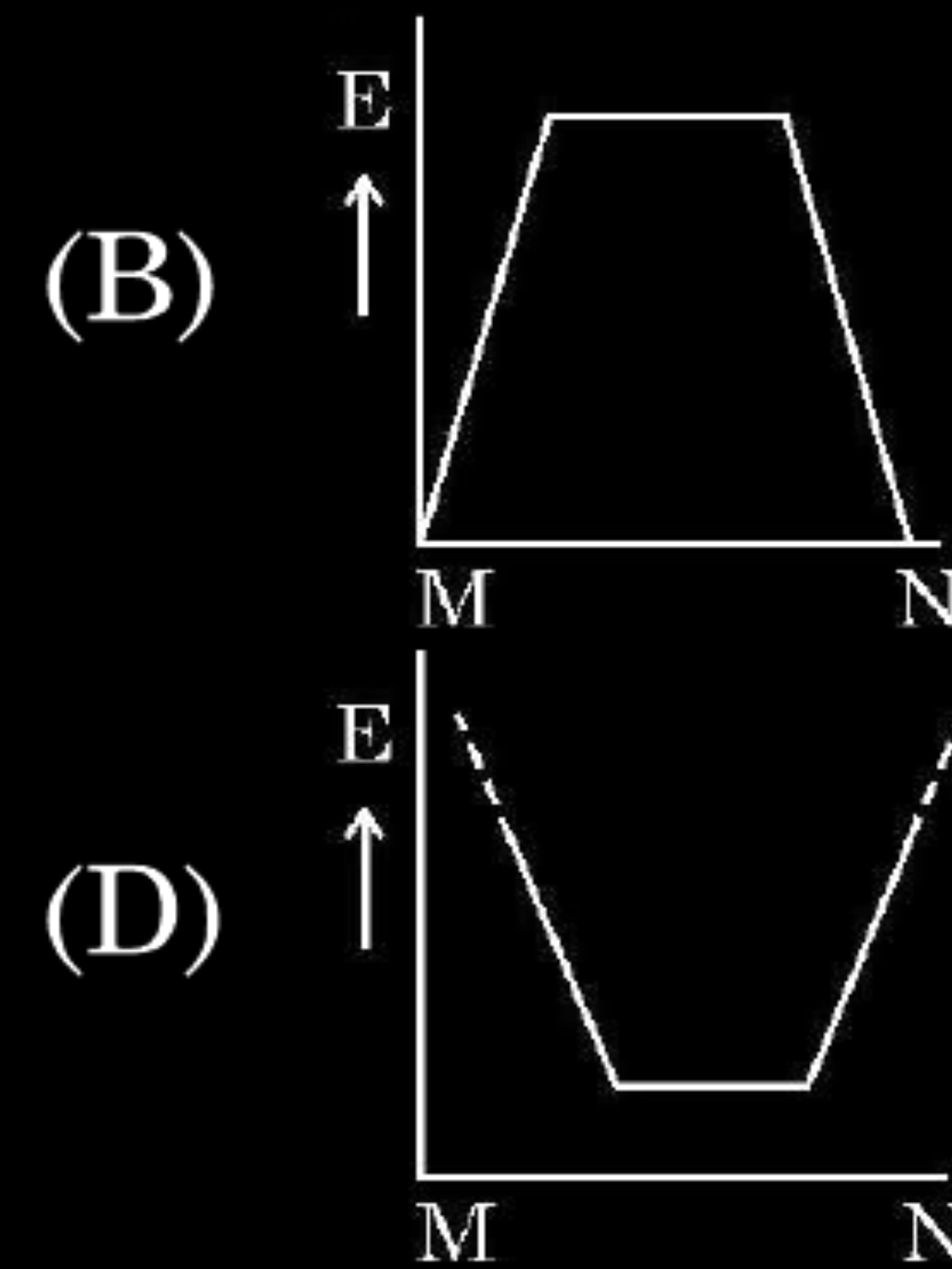
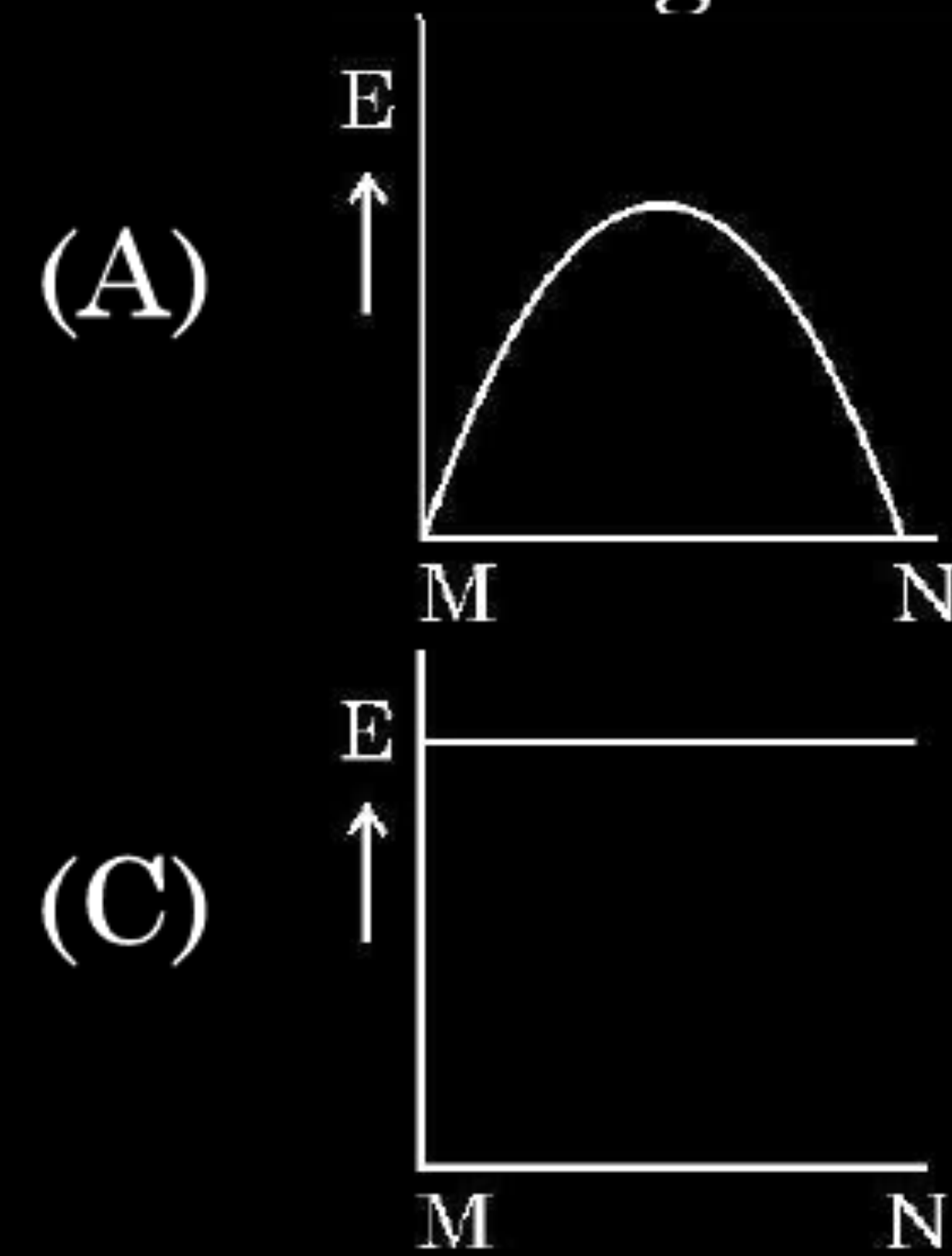
- (ii) A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which of the graphs correctly depict the potential difference (V) to be applied between the plates as a function of separation between the plates (d) to maintain the constant electric field ?



(iii)



In the above figure P, Q are the two parallel plates of a capacitor. Plate Q is at positive potential with respect to plate P. MN is an imaginary line drawn perpendicular to the plates. Which of the graphs shows correctly the variations of the magnitude of electric field strength E along the line MN?



(iv) Three parallel plates are placed above each other with equal displacement $\vec{d}$ between neighbouring plates. The electric field between the first pair of the plates is $\vec{E}_1$ and the electric field between the second pair of the plates is $\vec{E}_2$. The potential difference between the third and the first plate is –

(A) $(\vec{E}_1 + \vec{E}_2) \cdot \vec{d}$

(B) $(\vec{E}_1 - \vec{E}_2) \cdot \vec{d}$

(C) $(\vec{E}_2 - \vec{E}_1) \cdot \vec{d}$

(D) $\frac{d(E_1 + E_2)}{2}$

(iv) A material of dielectric constant K is filled in a parallel plate capacitor of capacitance C . The new value of its capacitance becomes

(A) C

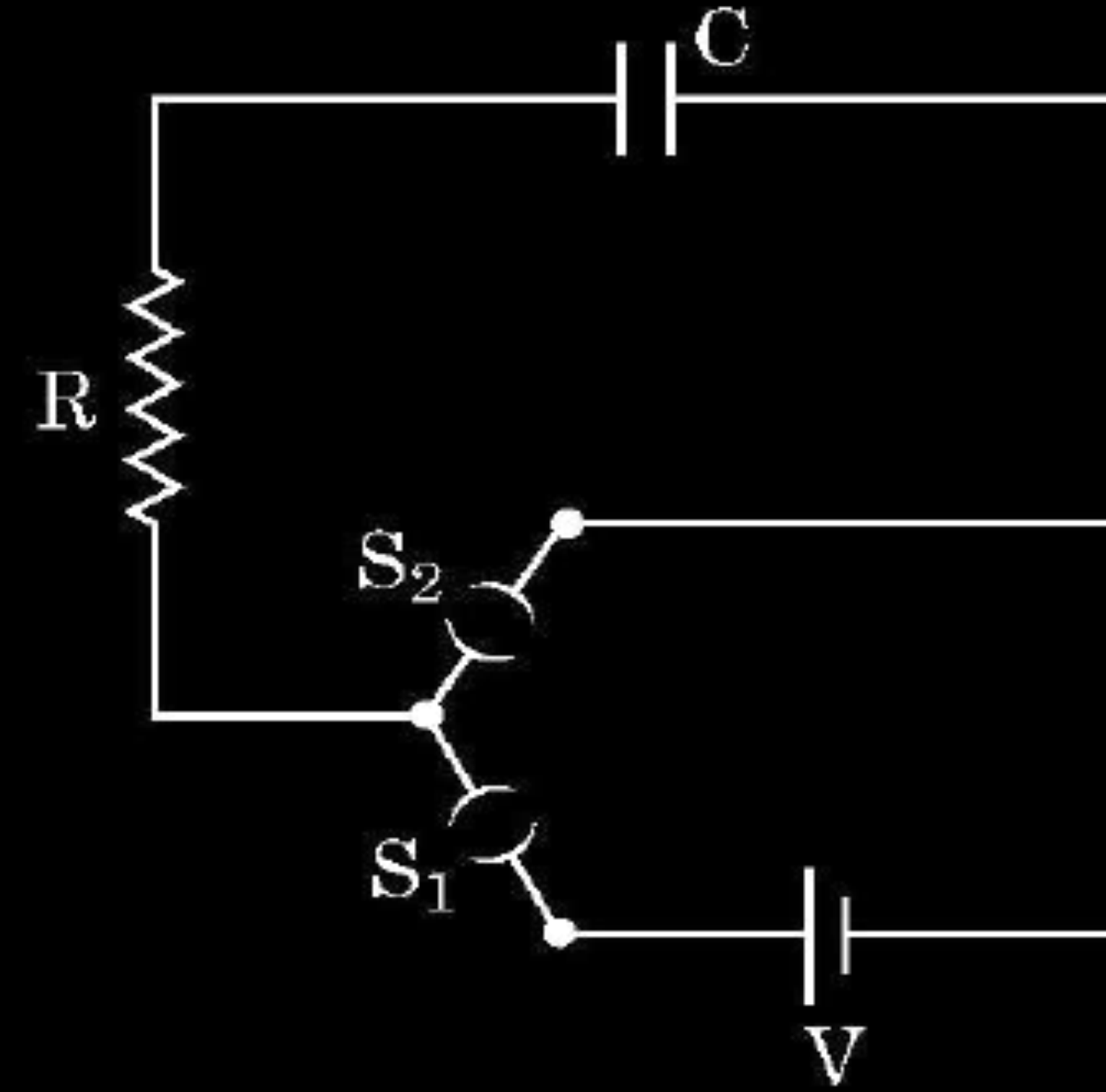
(B) $\frac{C}{K}$

(C) CK

(D) $C\left(1 + \frac{1}{K}\right)$

33. (a) (i) A small conducting sphere A of radius r charged to a potential V , is enclosed by a spherical conducting shell B of radius R . If A and B are connected by a thin wire, calculate the final potential on sphere A and shell B.
- (ii) Write two characteristics of equipotential surfaces. A uniform electric field of 50 NC^{-1} is set up in a region along $+x$ axis. If the potential at the origin $(0, 0)$ is 220 V , find the potential at a point $(4\text{m}, 3\text{m})$.

30. A circuit consisting of a capacitor C , a resistor of resistance R and an ideal battery of emf V , as shown in figure is known as RC series circuit. $4 \times 1 = 4$



As soon as the circuit is completed by closing key S_1 (keeping S_2 open) charges begin to flow between the capacitor plates and the battery terminals. The charge on the capacitor increases and consequently the potential difference $V_c (= q/C)$ across the capacitor also increases with time. When this potential difference equals the potential difference across the battery, the capacitor is fully charged ($Q = VC$). During this process of charging, the charge q on the capacitor changes with time t as $q = Q[1 - e^{-t/RC}]$

The charging current can be obtained by differentiating it and using $\frac{d}{dx}(e^{mx}) = me^{mx}$.

Consider the case when $R = 20 \text{ k}\Omega$, $C = 500 \text{ }\mu\text{F}$ and $V = 10 \text{ V}$.

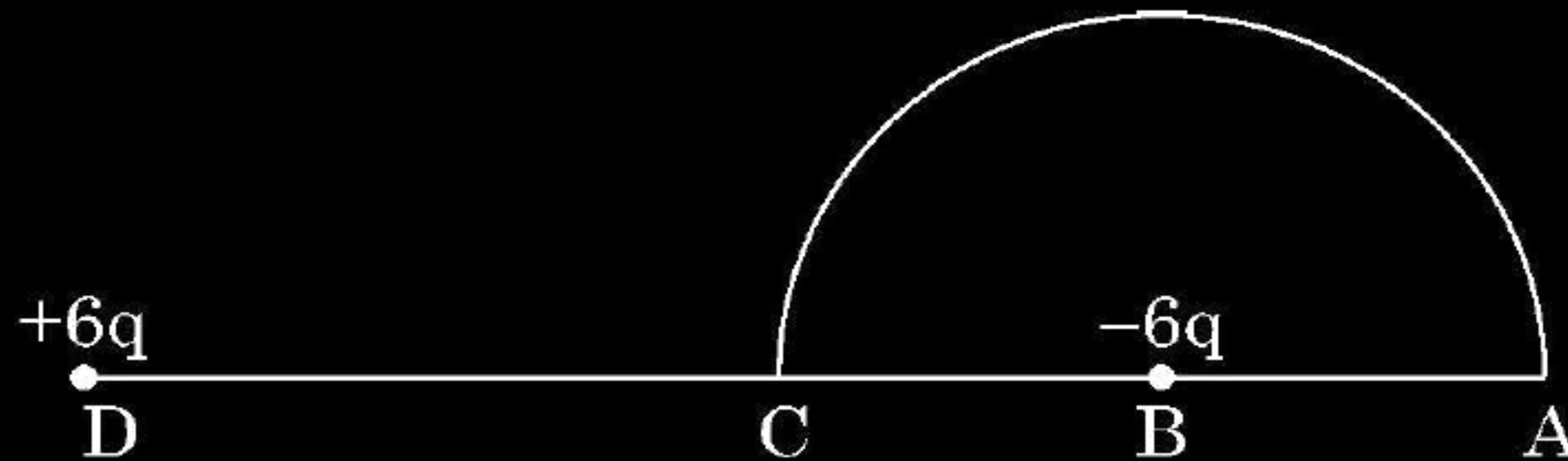
- (i) The final charge on the capacitor, when key S_1 is closed and S_2 is open, is
 (A) $5 \mu\text{C}$ (B) 5 mC
 (C) 25 mC (D) 0.1 C
- (ii) For sufficient time the key S_1 is closed and S_2 is open. Now key S_2 is closed and S_1 is open. What is the final charge on the capacitor ?
 (A) Zero (B) 5 mC
 (C) 2.5 mC (D) $5 \mu\text{C}$
- (iii) The dimensional formula for RC is
 (A) $[M L^2 T^{-3} A^{-2}]$ (B) $[M^0 L^0 T^{-1} A^0]$
 (C) $[M^{-1} L^{-2} T^4 A^2]$ (D) $[M^0 L^0 T A^0]$
- (iv) The key S_1 is closed and S_2 is open. The value of current in the resistor after 5 seconds, is
 (A) $\frac{1}{2\sqrt{e}} \text{ mA}$ (B) $\sqrt{e} \text{ mA}$ (C) $\frac{1}{\sqrt{e}} \text{ mA}$ (D) $\frac{1}{2e} \text{ mA}$

OR

- (iv) The key S_1 is closed and S_2 is open. The initial value of charging current in the resistor, is
 (A) 5 mA (B) 0.5 mA (C) 2 mA (D) 1 mA

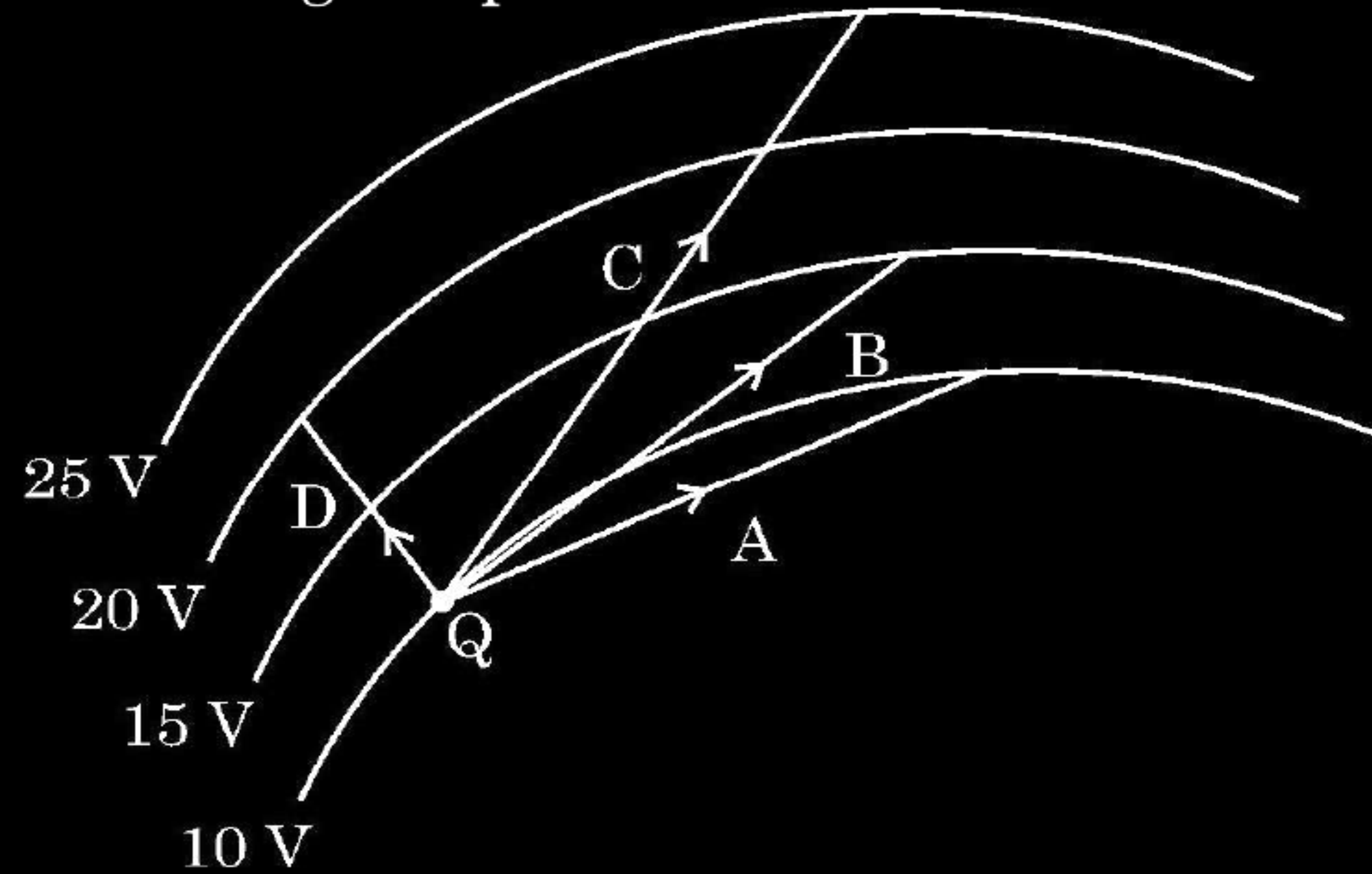
32. (a) (i) Two point charges $5\ \mu\text{C}$ and $-1\ \mu\text{C}$ are placed at points $(-3\ \text{cm}, 0, 0)$ and $(3\ \text{cm}, 0, 0)$ respectively. An external electric field $\vec{E} = \frac{A}{r^2} \hat{r}$ where $A = 3 \times 10^5\ \text{Vm}$ is switched on in the region. Calculate the change in electrostatic energy of the system due to the electric field.
- (ii) A system of two conductors is placed in air and they have net charge of $+80\ \mu\text{C}$ and $-80\ \mu\text{C}$ which causes a potential difference of $16\ \text{V}$ between them.
- (1) Find the capacitance of the system.
 - (2) If the air between the capacitor is replaced by a dielectric medium of dielectric constant 3, what will be the potential difference between the two conductors ?
 - (3) If the charges on two conductors are changed to $+160\ \mu\text{C}$ and $-160\ \mu\text{C}$, will the capacitance of the system change ? Give reason for your answer.

- (b) (i) Consider three metal spherical shells A, B and C, each of radius R . Each shell is having a concentric metal ball of radius $R/10$. The spherical shells A, B and C are given charges $+6q$, $-4q$, and $14q$ respectively. Their inner metal balls are also given charges $-2q$, $+8q$ and $-10q$ respectively. Compare the magnitude of the electric fields due to shells A, B and C at a distance $3R$ from their centres.
- (ii) A charge $-6 \mu\text{C}$ is placed at the centre B of a semicircle of radius 5 cm, as shown in the figure. An equal and opposite charge is placed at point D at a distance of 10 cm from B. A charge $+5 \mu\text{C}$ is moved from point 'C' to point 'A' along the circumference. Calculate the work done on the charge.



1. In the figure curved lines represent equipotential surfaces. A charge Q is moved along different paths A, B, C and D. The work done on the charge will be maximum along the path

1



(A) A
(C) C

(B) B
(D) D